



西南交通大学
Southwest Jiaotong University

S RTP开题答辩 ▶

基于用户需求预测的 共享单车再平衡

答辩学生：蒋昕



目录

CONTENTS

1. 项目背景
2. 项目内容
3. 项目创新性
4. 项目基础
5. 项目计划

第一部分 ▶

项目背景

- 研究背景
- 研究意义



新华社北京11月2日电 习近平同志在党的二十大报告中强调：“必须牢固树立和践行绿水青山就是金山银山的理念，站在人与自然和谐共生的高度谋划发展。”这是立足我国进入全面建设社会主义现代化国家、实现第二个百年奋斗目标的新发展阶段，对推动经济社会发展提出的新要求。学习贯彻党的二十大精神，就要深刻把握生态文明建设这个关乎中华民族永续发展的根本大计，扎实推进绿色发展，促进人与自然和谐共生，共同建设美丽中国。

新华社评论员：推动绿色发展，建设美丽中国——七论学习贯彻党的二十大精神

新华社北京11月2日电 习近平同志在党的二十大报告中强调：“必须牢固树立和践行绿水青山就是金山银山的理念，站在人与自然和谐共生的高度谋划发展。”这是立足我国进入全面建设社会主义现代化国家、实现第二个百年奋斗目标的新发展阶段，对推动经济社会发展提出的新要求。学习贯彻党的二十大精神，就要深刻把握生态文明建设这个关乎中华民族永续发展的根本大计，扎实推进绿色发展，促进人与自然和谐共生，共同建设美丽中国。

新华社北京11月2日电 习近平同志在党的二十大报告中强调：“必须牢固树立和践行绿水青山就是金山银山的理念，站在人与自然和谐共生的高度谋划发展。”这是立足我国进入全面建设社会主义现代化国家、实现第二个百年奋斗目标的新发展阶段，对推动经济社会发展提出的新要求。学习贯彻党的二十大精神，就要深刻把握生态文明建设这个关乎中华民族永续发展的根本大计，扎实推进绿色发展，促进人与自然和谐共生，共同建设美丽中国。

良好生态环境是公平的公共产品，是最普惠的民生福祉。全国地级及以上城市优良天数比率增加8.75%，单位国内生产总值能耗下降26.4%，万元工业增加值用水量下降53%……党的十八大以来，以习近平同志为核心的党中央坚持把生态文明建设摆在突出位置，全方位、全地域、全过程加强生态环境保护，经过不懈努力，生态文明制度体系更加健全，污染防治攻坚战深入推进，绿色、低碳、循环、集约、安全、韧性、普惠的生态文明制度体系更加健全，生态文明建设取得重大成就，生态环境明显改善，绿色发展水平显著提升，生态文明建设取得重大成就，生态环境明显改善，绿色发展水平显著提升。

中国
社会主义现
代化建设
中，应对



研究背景

在党的二十大报告提出的加快发展方式绿色转型目标，新型城镇化战略和碳达峰、碳中和目标的指导下，共享单车为解决“最后一公里”的便携出行问题提供了经济、低碳、环保的有效策略。共享单车市场的持续扩大不仅有助于缓解城市交通压力和降低环境污染，还是落实国家重大战略目标的具体举措，对推动城市可持续发展具有重要意义。

研究意义 ▶

随着共享单车市场迅速扩张，随意停放影响交通秩序，车辆损坏严重等问题随之暴露；根据用户需求预测进行重平衡可以有效节省人力物力，实现放管结合，助力共享单车健康发展。



第二部分 ▶

项目内容

- 需求预测
- 再平衡优化

需求预测

Expansion Plan Data

Capital Bikeshare conducted a development plan survey every few years to assess usage trends and [expansion plans](#). The proposed expansion plan is available to download as a [KML file](#).

Member Surveys

Capital Bikeshare conducts surveys of its member communities. The results are available on the [Capital Bikeshare website](#).

2016

[2016 Capital Bikeshare Member Survey](#)

[2015 Capital Bikeshare Member Survey](#)

2014

[2014 Capital Bikeshare Member Survey](#)

[2014 Capital Bikeshare Member Survey](#)

2012

[2012 Capital Bikeshare Member Survey](#)

[2012 Capital Bikeshare Member Survey](#)

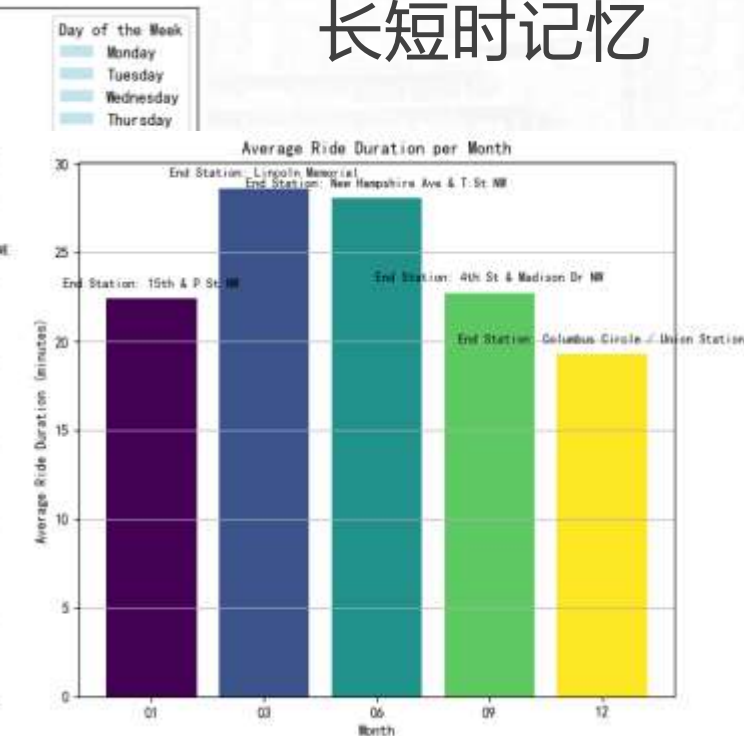
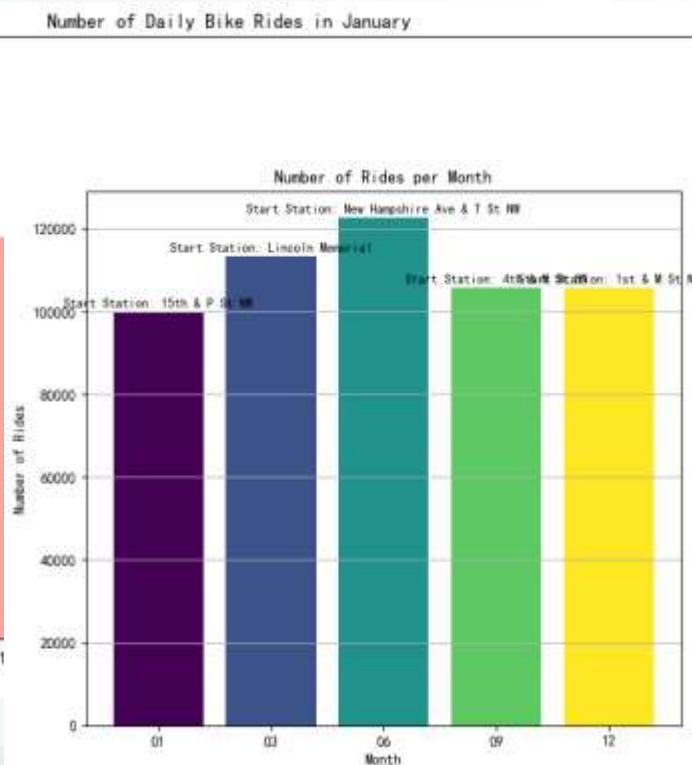
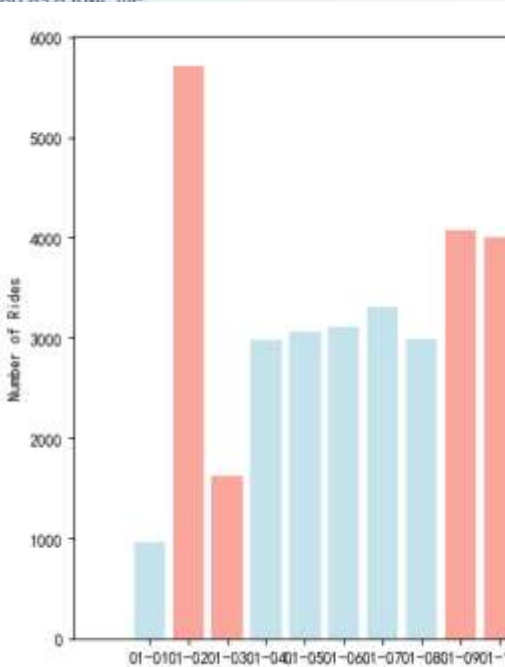
[Health Implications of Capital Bikeshare](#)

2011

[2011 Capital Bikeshare Member Survey](#)

[2011 Capital Bikeshare Survey](#)

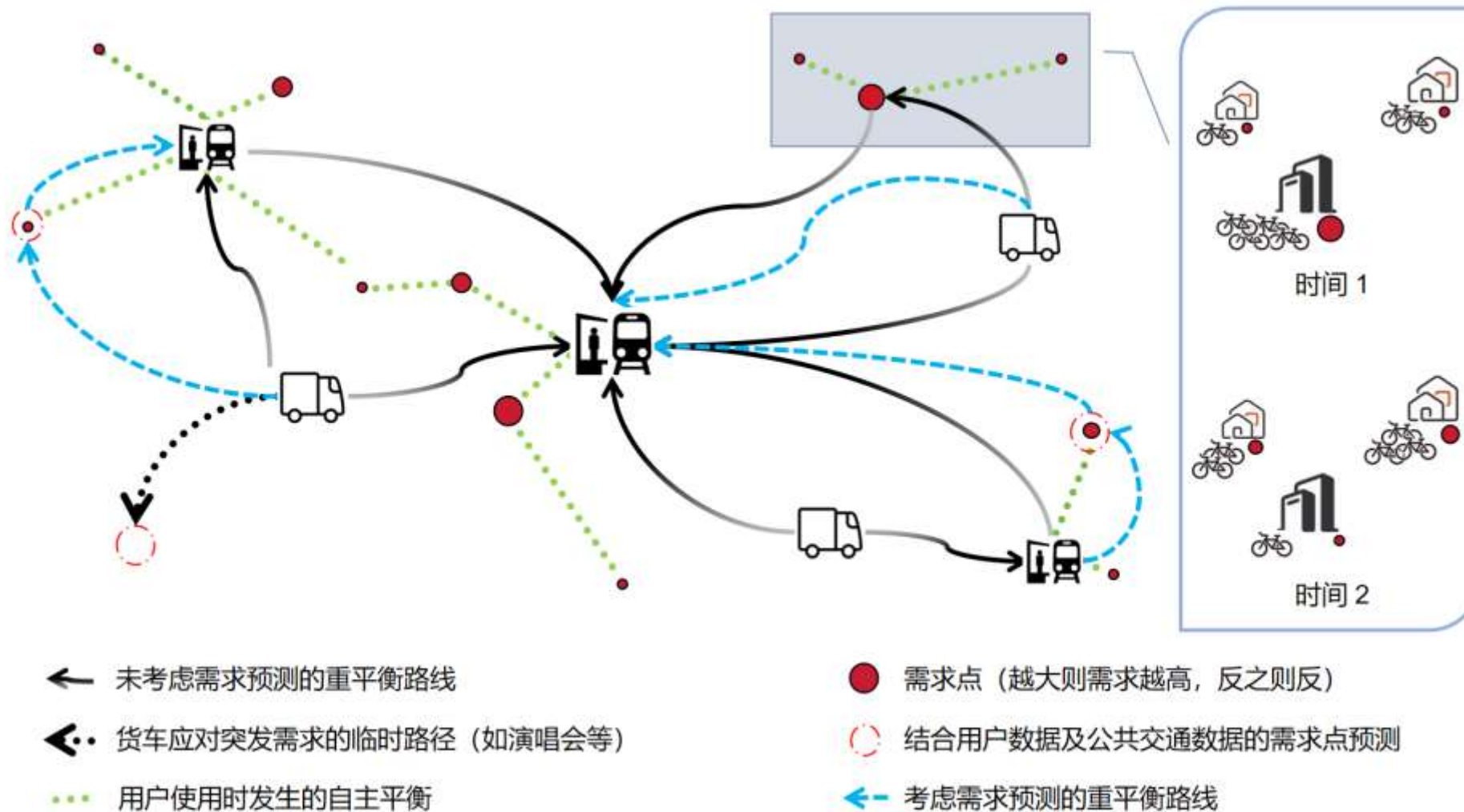
Updated on December 6, 2016.



随机森林
长短时记忆

再平衡优化

混合整数规划
自适应大邻域算法

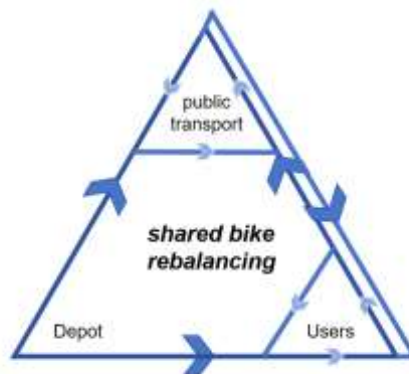


第三部分 ▶

项目创新性

- 前人不足
- 创新点

前人不足 ▶



总之，共享单车再平衡问题近年来引起了许多研究人员的兴趣。大部分研究集中于解决问题本身，而没有考虑用户行为及其在再平衡问题中的作用。而且，大多数研究都是在较小的网络上进行实验，文献大多将网络简化为矢量图并进行调查。很少有研究使用聚类降低大规模优化问题的复杂性 [8,10,13,15,23]，而大多数再平衡研究都没有考虑共享单车系统中的用户行为。用户行为是需求和供给之间内在互动的结果。自行车的可用性（供给）距离和地理地形会影响用户行为，进而影响对自行车和停靠点的需求。通过考虑站点之间的行程对网络进行聚类，可以使网络分析变得更简单、更高效，而且正如下文将讨论的那样可以消除不必要的行程。

忽略部分

1. 进行需求预测时仅根据运营商历史数据作为依据，忽略部分地区因为共享单车数量少而放弃使用共享单车这一现象，陷入恶性循环

较为笼统

2. 预测范围大但较为笼统，无法为共享单车的资源分配提供更加细分的数据支持

分类简单

3. 预测时考虑因素较少，预测时的时间分类较为简单，不适用于实际应用

前人不足

模式单一

4.采用单一静态再平衡，不适配于复杂交通场景。

References	Year	Dynamic	HBSS	Objective	Model	Solution Method	Dataset
Lin et al. [5]	2018	×	✓	(1) the total number of the unsatisfied customers (2) the inconvenience of getting a bike from the HBSS (3) weighted vehicles' total operational time	MP	chemical reaction optimization	data from New York, NY, USA
Chen et al. [6]	2013	×	×	travel cost	MP	branch-and-cut	randomly generated instances
Zhang et al. [11]	2015	×	✓	(1) the upper level: the repositioning workload (2) the lower level: the lower profit the user profit	MP	a bi-level programming model	data from Philadelphia, PA, USA
Li et al. [16]	2010	×	×	total travel cost	MP	memetic algorithm	30 real-world instances
Ren et al. [17]	2019	×	×	depot inventory cost and traveling cost	MP	branch-and-cut	random demand datasets and a real dataset that operates at the University of South Florida
Mahmoudian et al. [18]	2012	×	✓	level of service and rebalancing cost	multi-objective optimization	multi-objective genetic algorithm	

算法局限

5.采用遗传算法等启发式算法，使得优化结果易于陷入局部最优解。

References	Year	Dynamic	HBSS	Objective	Model	Solution Method	Dataset
Chen et al. [5]	2018	✓	✓	(1) the total cost of the rebalancing vehicle and the degree of imbalance in the HBSS	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS
Rego and Becker [19]	2014	✓	×	Inventory levels, travel cost and customer learning	Inventory levels, travel cost and customer learning	genetic algorithm	data from the European Distribution network
Shin and Tseng [21]	2011	✓	×	the rebalancing vehicle cost	the rebalancing vehicle cost	genetic algorithm	data from the European Distribution network
Salazar et al. [22]	2017	×	×	repositioning	MP	Matlab	data from the European Distribution network
Beck et al. [23]	2011	×	×	travel cost	graph theory	approximation algorithm	data from the European Distribution network
He and Li [24]	2014	×	×	total profit	MP	genetic algorithm	data from the European Distribution network
Li et al. [16]	2010	×	×	total cost	MP	genetic algorithm	data from the European Distribution network
Lin et al. [11]	2010	×	×	rebalancing cost	MP	genetic algorithm	data from the European Distribution network
Ren et al. [17]	2019	×	×	total cost	MP	branch-and-cut	data from the European Distribution network
Li et al. [16]	2010	×	×	rebalancing cost	MP	genetic algorithm	data from the European Distribution network
Chen et al. [5]	2018	✓	✓	(1) the degree of imbalance in the HBSS (2) the greenhouse gas emissions	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS
Chen et al. [5]	2018	✓	✓	(1) the degree of imbalance in the HBSS (2) the greenhouse gas emissions	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS
Chen et al. [5]	2018	✓	✓	(1) the degree of imbalance in the HBSS (2) the greenhouse gas emissions	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS

目标单一

6.优化目标较为单一，仅将再平衡所需里程数及成本作为目标，忽略碳排放等潜在目标，不符合绿色出行主旨。

References	Year	Dynamic	HBSS	Objective	Model	Solution Method	Dataset
Chen et al. [5]	2018	✓	✓	(1) the total cost of the rebalancing vehicle and the degree of imbalance in the HBSS	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS
Rego and Becker [19]	2014	✓	×	Inventory levels, travel cost and customer learning	Inventory levels, travel cost and customer learning	genetic algorithm	data from the European Distribution network
Shin and Tseng [21]	2011	✓	×	the rebalancing vehicle cost	the rebalancing vehicle cost	genetic algorithm	data from the European Distribution network
Salazar et al. [22]	2017	×	×	repositioning	MP	Matlab	data from the European Distribution network
Beck et al. [23]	2011	×	×	travel cost	graph theory	approximation algorithm	data from the European Distribution network
He and Li [24]	2014	×	×	total profit	MP	genetic algorithm	data from the European Distribution network
Li et al. [16]	2010	×	×	total cost	MP	genetic algorithm	data from the European Distribution network
Lin et al. [11]	2010	×	×	rebalancing cost	MP	genetic algorithm	data from the European Distribution network
Ren et al. [17]	2019	×	×	total cost	MP	branch-and-cut	data from the European Distribution network
Li et al. [16]	2010	×	×	rebalancing cost	MP	genetic algorithm	data from the European Distribution network
Chen et al. [5]	2018	✓	✓	(1) the degree of imbalance in the HBSS (2) the greenhouse gas emissions	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS
Chen et al. [5]	2018	✓	✓	(1) the degree of imbalance in the HBSS (2) the greenhouse gas emissions	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS
Chen et al. [5]	2018	✓	✓	(1) the degree of imbalance in the HBSS (2) the greenhouse gas emissions	the rebalancing vehicle theory	solving heuristic strategy	data from the Shanghai HBSS

目标，包括最大限度地减少车辆行驶时间和最大限度地减少交通流量[10]。
成本、行程长度、汽车尾气排放，以及最大限度地减少用户的不满和未满足的需求。

表 1.再平衡文献摘要。

参考资料	类型	数学方法	目标
Dell'Amico 等人 [1]	静态	混合整数线性规划	尽量减少旅行总费用
克鲁兹等人 [11]	静态	迭代局部搜索 (ILS)	尽量减少旅行总费用
Chen et al. [14]	静态	整数程序	尽量减少旅行总费用
Forma 等人 [10]	静态	混合整数线性规划	最大限度地减少未获得服务的用户数量和总旅行距离
Lin 等人 [15]	静态	混合整数线性规划	最大限度地减少总行程
Schuijbroek 等人 [16]	静态	约束编程	最佳车辆路线
李等人 [4]	静态	混合整数线性规划	将总成本降至最低
Alvarez-Valdes 等人 [12]	静态	一种启发式算法	将未满足需求的总体成本降至最低
Pal 和 Zhang [3]	静态	混合整数线性规划	尽量缩短再平衡车队的生产周期
埃多斯等人 [17]	静态	精确法	最小化再平衡成本
埃多斯等人 [18]	静态	精确法整数编程	最大限度地减少差旅费和手续费
Shui 和 Szeto [9]	动态	人工蜂群算法	最小化未满足的总需求以及燃料和 CO ₂ 排放成本
Zhang 等人 [6]	动态	混合整数问题	最大限度地减少未满足的总需求

1.需求预测与再平衡算法的整合

项目将开发一个结合实时需求预测和动态再平衡的综合性算法，以实现更精确的车辆调度。

3.环境可持续性

项目特别关注再平衡操作的环境影响，旨在通过减少空驶和提高车辆利用率来降低碳排放。



2.城市交通网络的实证分析

项目将通过结合某区域下不同城市交通的数据，辅助共享单车运营商历史数据预测该区域共享单车需求并进行再分配。

4.共享单车使用率提升

通过优化再平衡策略，项目预期将提高用户满意度和微出行工具的使用频率。

第四部分 ▶

项目基础

- 现有进展
- 成员介绍



现有进展▶

文献综述

2. Literature Review

2.1. Complexity of Rebalancing Problem

The complexity of the rebalancing problem in bike-sharing systems is discussed in Zhang et al. (2023). Bike-sharing systems rebalancing considering redistribution proportions: A user-based repositioning approach. The paper proposes solutions such as determining optimal vehicle routes and the quantity of bikes to place or retrieve. Differences between BSSs, such as variations in bike types and relocation strategies, contribute to the complexity. A two-level planning model is introduced, and user-based relocation methods are explored.

2.2. Static Bike Sharing Rebalancing Problem (SBRP)

Chuetre and Andrade (2023) introduces the Static Bike Sharing Rebalancing Problem (SBRP), aiming to maintain system stability through bike

Preprint submitted to Nuclear Physics B May 20, 2024

redistribution. Both truck-based and user-based relocation methods are analysed, and a nonlinear Mixed Integer Programming (MIP) formula and a metaheuristic algorithm (IGVNS) are proposed to address SBRP.

2.3. Impact of Imbalance on System Operation

Hackkrishnakumar et al. (2022) addresses the impact of imbalance on system operation and discusses the complexity of solutions, including determining optimal vehicle routes and crew-related constraints. A two-level planning model is proposed, with a study conducted on the static BRP of Lisbon's BSS-GIRA.

2.4. Role of BSS in Urban Transportation

Liang et al. (2024) discusses the role of Bike Sharing Systems (BSS) in urban transportation and environmental challenges. It highlights the importance of decision support frameworks based on simulation models and machine learning algorithms. The advantages of data-driven approaches are emphasized, and a novel simulation and machine learning-based method are introduced to expedite BSS decision-making processes and tackle rebalancing issues.

2.5. Conclusion

The literature reviewed highlights the importance of effective rebalancing strategies in bike-sharing systems for maintaining system stability and customer satisfaction. Various approaches, including optimization models, metaheuristic algorithms, and machine learning techniques, have been proposed to address the rebalancing problem and improve BSS operation. Further research in this area is needed to devel

2.6. Dynamic repositioning in bike-sharing systems with uncertain demand: An improved rolling horizon framework

Li et al. (2023) proposes a multi-period two-stage stochastic model based on the spatiotemporal imbalance of BSS caused by fluctuating user demand. Through studying the dynamic bike repositioning problem under scenario-based demand, which incorporates a vehicle no-return (VNR) strategy that allows vehicles to remain at the last station at the end of each period, awaiting the next repositioning task. To solve this model, the article introduces a customized hybrid heuristic algorithm based on parallel genetic algorithm and variable neighborhood search.

2.7. Spatial Interaction Analysis of Shared Bicycles Mobility Regularity and Determinants: A Case Study of Six Main Districts, Beijing

Hu et al. (2022) emphasizes the significant of understanding the regularity and determinants of mobility when taking care of the reasonable deployment of shared bicycles and urban planning. It creates a spatial interaction network based on the data from bike sharing data from six main districts in Beijing and use the exponential random graph model (ERGM) to interpret the influencing factors of the network structure and the mobility regularity. As a result, it obtains some characteristics of the spatial interaction network structure between weekdays and weekends. Additionally, ERGM shows that the influencing factors are positively correlated with the spatial interaction network, where three determinants including building density, housing prices and the number of residential areas matters most.

2.8. A two-stage robust approach to integrated station location and rebalancing vehicle service design in bike-sharing systems

Fu et al. (2022) This article proposes a new integrated station location and rebalancing vehicle service design model, which is targeted at maximizing daily revenue under a given total investment for station locations and bike acquisition. Then it presents a two-stage robust optimization model with a demand-related uncertainty set. The first stage of model determines the station locations, initial bike inventory, and service areas of rebalancing vehicles. After that, it describe the problem as an assignment of each rebalancing vehicle to a service area composed of several specified stations. In the second stage, its model optimizes the daily fleet operation and maximizes the total revenue minus the rebalancing cost. A customized row generation approach is developed to solve the model. Compared with stochastic models with small data set, this robust model can achieve larger mean and worst-case revenues and a higher revenue stability.

2.9. An improved general variable neighborhood search for a static bike-sharing rebalancing problem considering the depot inventory

Ren et al. (2020) This article researches the integration of the depot inventory and vehicle routing problems, which is rarely considered in former study. By proposing two mixed integer programming (MIP) formulations to find the daily optimal decision on the vehicle routes and the numbers of bikes and vehicles employed from the depot, it lays a solid foundation for the

Table 1. Comparison between the proposed model and existing approaches in the literature.

Article	Problem characteristics				Methodologies			
	Mode	Problem	routing	Uncertainty	location	Approach	Learning	Re-planning
Synchronodal transport (Zhang et al. (2022)	road, railway, inland waterway	STP	-	-	-	RHC	-	periodical

- in the "Required prior information" column, it means that no information is required as uncertainty is not taken into account, in other columns, it means that the relevant item is not mentioned in the article.
RHC: Rolling horizon control; STP: Synchronodal transport planning; VRP: Vehicle Routing Problem; MDP: Markov decision process; DT: Decision trees; LP: Linear programming model; SO: Stochastic optimization.

现有进展 ▶ 代码学习与复现

部分代码截图

```
# Is GPU available
deviceIDs = GPUtil.getAvailable(orders='last', limit=8, maxLoad=1, maxMemory=0.2,
                                includeNan=False, excludeID=[], excludeUUID=[])

if len(deviceIDs) == 8:
    current_device = '-1'
else:
    current_device = str(deviceIDs[0])
DEVICE = 'cuda:{}'.format(current_device)

#parser
import warnings
args = argparse.ArgumentParser(description='Argument Parser')
args.add_argument('--dataset', default='', type=str)
args.add_argument('--city', default=None)
args.add_argument('--MergeIndex', default=3)
args.add_argument('--MergeWay', default='sum', type=str)
args.add_argument('--test_ratio', default=0.1, type=float)

#data source
DATASETPATH =

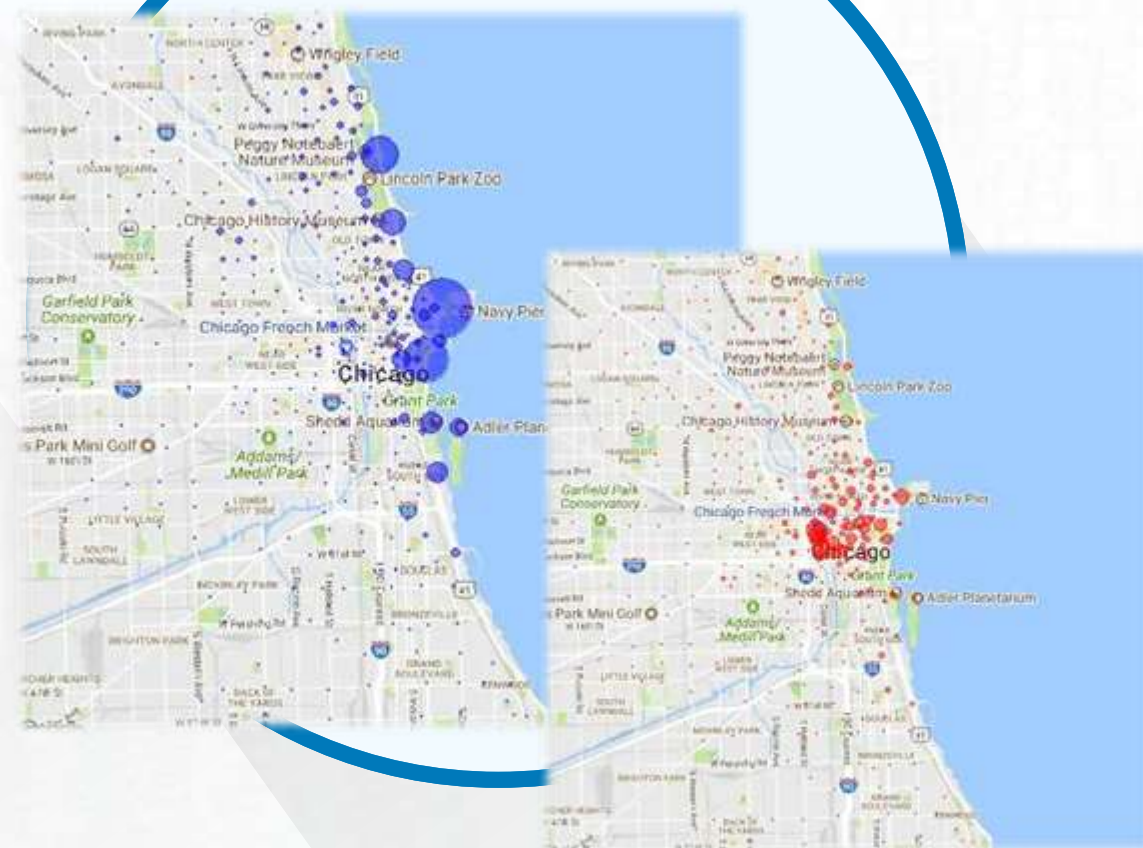
#get config
config = config
config.read(D

#device
args.add_argument(
    parser.add_argument( 'name_or_flag: "--CT", default=168, type=int)

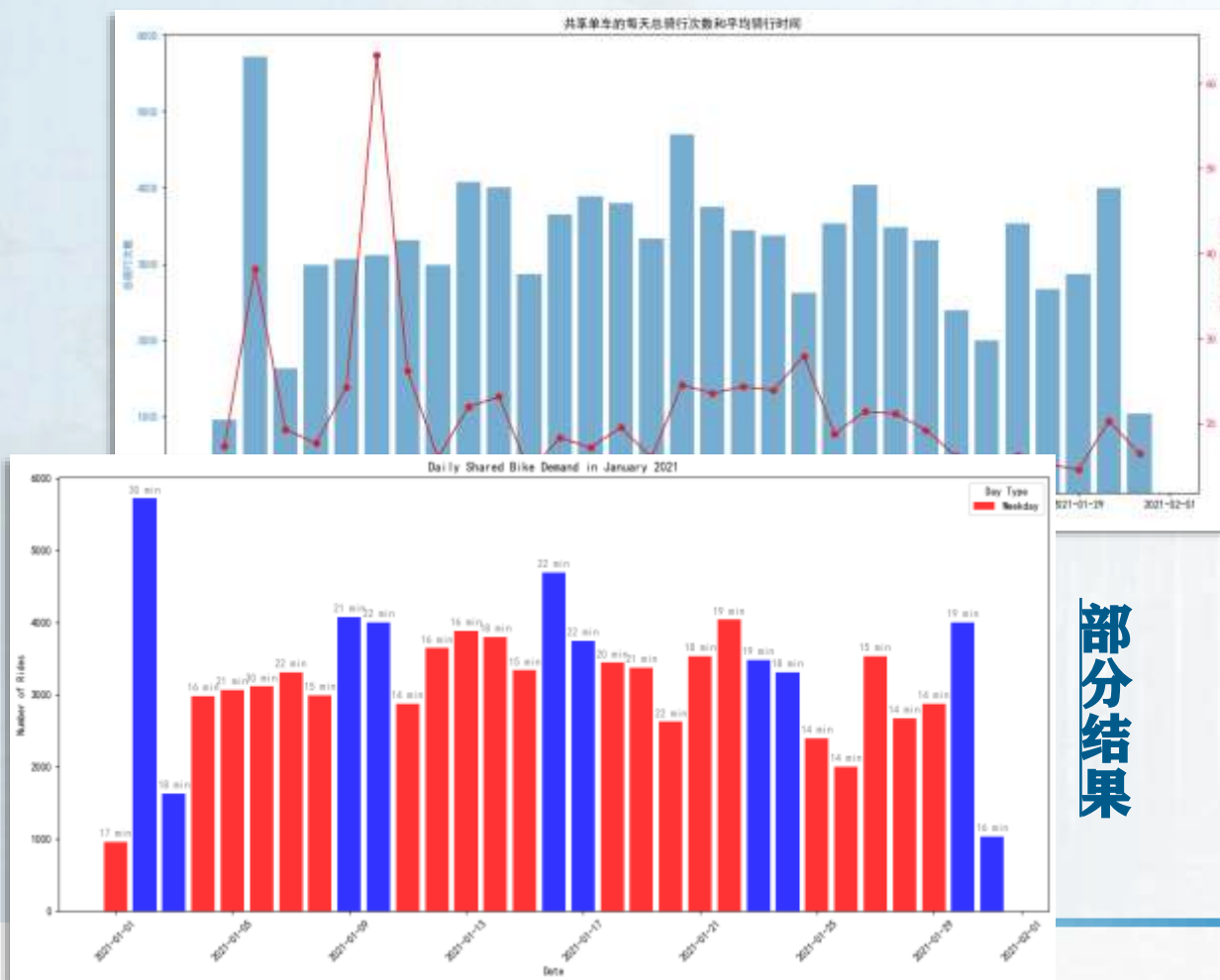
    parser.add_argument( 'name_or_flag: "--ar", default=3, type=int)
    parser.add_argument( 'name_or_flag: "--d", default=8, type=int)
    parser.add_argument( 'name_or_flag: "--na", default=1, type=int)

    parser.add_argument( 'name_or_flag: "--sa", default=0, type=int)
    parser.add_argument( 'name_or_flag: "--sd", default=0, type=int)
    parser.add_argument( 'name_or_flag: "--sna", default=8, type=int)
    parser.add_argument( 'name_or_flag: "--sp", default=8, type=int)
```

部分复现结果



现有进展 ▶ 数据可视化



```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 6))

plt.bar(monthly_stats['Month'], monthly_stats['Number of Rides'], width=0.4, label='Number of Rides')

plt.plot(monthly_stats['Month'], monthly_stats['Average Ride Duration (minutes)'], label='Average Ride Duration (minutes)')

plt.xlabel('Month')
plt.ylabel('Value')
plt.title('Comparison of Number of Rides and Average Ride Duration for Different Months')
plt.xticks(rotation=45)
plt.legend()
```

```
plt.figure(figsize=(12, 6))

for day, group in rides_per_day.groupby('day_of_week'):
    if day in ['Saturday', 'Sunday']:
        color = milder_weekend_color
    else:
        color = milder_weekday_color
    plt.bar(group['date'], group['rides'], label=day, alpha=0.7, color=color)

plt.gca().set_xticks(rides_per_day['date'])
plt.gca().set_xticklabels(rides_per_day['date'].dt.strftime('%m-%d'))
plt.xlabel('Date (mm-dd)')
plt.ylabel('Number of Rides')
plt.title('Number of Daily Bike Rides in January')
plt.legend(title='Day of the Week')
plt.tight_layout()
```

部分代码截图

部分结果

成员介绍▶

杨晨杰

数学建模新秀杯并获得三等奖，校赛获得二等奖；熟练掌握Python等编程语言；参加了美国大学生数学建模竞赛和ACM校队选拔赛。

蒋昕

数学建模新秀杯并获得三等奖，校赛获得二等奖；掌握Python、C++等语言；熟练使用SPSS、LaTeX、MATLAB等软件。

王婧琪

建筑类专业第三；优秀共青团员；熟练使用SPSS、SU、CAD、Procreate等软件。

第五部分 ▶

项目计划

- 进度安排
- 预期结果

进度安排 ▶



预期结果

- 开发一个结合实时需求预测和动态再平衡的综合性算法。
- 优化微出行工具的再平衡，实现公共交通与共享服务的无缝连接。
- 构建共享单车重平衡需求预测准确度验证模型。



- 1、申请发明专利1项;
- 2、撰写文献综述;
- 3、撰写学术论文1篇。



谢谢观看
▼
敬请各位老师批评指正

答辩学生：蒋昕